Cosmic Watch: a live, explorable observatory that applies computer vision and machine learning to public astronomy data streams

Jay Gupta[1]

[1] SVKM's NMIMS Mukesh Patel School of Technology Management & Engineering, Mumbai, India

Preview render · 11 November 2026 · Software: github.com/jay4linkh/cosmic-watch-ai

Summary

`Cosmic Watch` is an open-source Python and WebGL application that continuously collects newly released public data from observatories and spacecraft, analyses it with computer vision and trained neural networks, checks the results against archival catalogues, and presents everything in one interactive 3D environment (Figure 1). Its thirteen live feeds include Rubin (Ivezić et al., 2019) and ZTF (Bellm et al., 2019) alerts via the ALeRCE broker (Förster et al., 2021), new JWST and Hubble images from MAST, SDO solar images (Pesnell et al., 2012; Lemen et al., 2012), SOHO/LASCO coronagraph data (Brueckner et al., 1995), NOAA space weather, gravitational-wave events, NASA GCN circulars, Mars rover images and NASA/JPL near-Earth-object data.

Deep-field sources are extracted with `SEP` (Barbary, 2016; Bertin & Arnouts, 1996) and classified by a galaxy-morphology CNN fine-tuned from `Zoobot` (Walmsley et al., 2023); solar images are searched for sunspots, active regions and coronal holes; rover images are classified by a terrain CNN; and detections are cross-matched against Gaia DR3 (Gaia Collaboration et al., 2023), Pan-STARRS1 (Chambers et al., 2016), SDSS, the Legacy Surveys (Dey et al., 2019) and SIMBAD (Wenger et al., 2000), with every claim stored alongside its evidence. Users can open any detection in a tiled deep-zoom view of the real pixels and label it, and those labels drive an active-learning loop that promotes a retrained model only when it passes held-out checks.

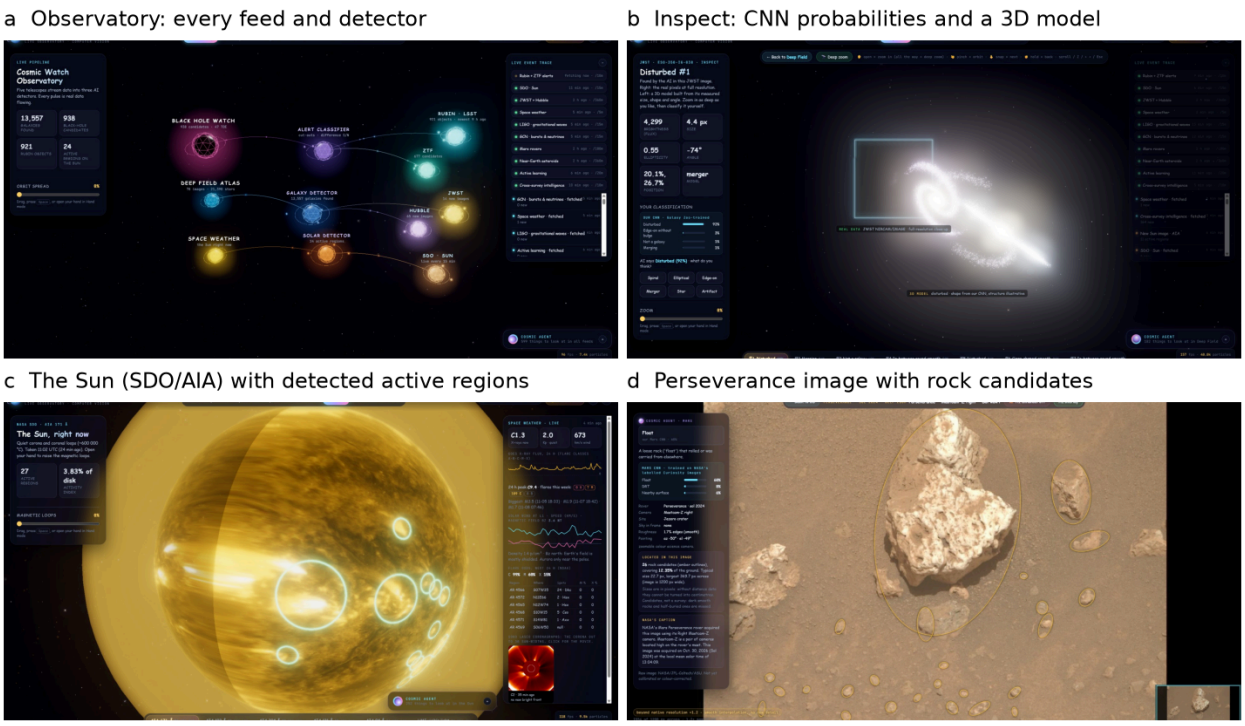


Figure 1: The running application, from a live screen recording. (a) Observatory overview: each feed and detector is a node. (b) Inspect mode for a JWST detection: CNN probabilities beside the real pixels. (c) SDO/AIA 171 Å with detected active regions and live NOAA space weather. (d) A Perseverance image classified by the Mars CNN, with rock candidates outlined.

Statement of need

Most of this data is free and public within minutes to days of being taken, but it is spread across many archives, formats and APIs. A student, educator or amateur astronomer who wants to follow the sky (what JWST released today, whether the Sun is flaring, which transients Rubin flagged last night) must visit many services and has no easy way to run modern analysis on what they find. Brokers and archive tools excel at their own data type but are not built to put many data types side by side or to explain a result to a non-specialist.

`Cosmic Watch` provides one reproducible pipeline that runs on a laptop with a consumer GPU, needs no API keys, and keeps a clear line between measurement and model output; for example, super-resolved pixels from SwinIR (Liang et al., 2021) or Real-ESRGAN (Wang et al., 2021) are always labelled "invented detail" and never reach a detector or classifier. It is aimed at teaching and outreach, student and citizen-science projects that need a readable end-to-end example, and quick-look triage of new releases and alerts.

Design

A FastAPI server and a scheduled pipeline write to SQLite; each source has a fetcher and each data type a detector. On calibrated JWST/HST mosaics, detection runs in several passes (cold, hot and binned low-surface-brightness), and each pass is also run on the sign-flipped image, where any detection is noise; thresholds rise until that spurious rate is below 3 %, and completeness is measured by injecting synthetic galaxies. The galaxy classifier is a Zoobot ConvNeXt-nano (Liu et al., 2022) pre-trained on Galaxy Zoo (Walmsley et al., 2022) and fine-tuned on Galaxy10 DECaLS (Leung & Bovy, 2019), with a ResNet-18 (He et al., 2016) baseline. Active learning adds user labels, weak labels and an eleventh "not a galaxy" class, and a challenger replaces the champion only if Galaxy10 accuracy falls by at most 1.5 points, held-out stars and gas are rejected with at least 60 % recall, and at most 10 % of held-out galaxies are rejected. An Isolation Forest (Liu et al., 2008) scores deep-field outliers, and a Thompson-sampling

bandit (Thompson, 1933) ranks findings for the user. Cross-matches use a Bayesian likelihood ratio of separation against local source density, with Gaia proper motions applied. Rover images are classified by a ConvNeXt-Tiny trained on Lu & Wagstaff (2020). The front end uses Three.js and OpenSeadragon, with optional in-browser hand-gesture control through MediaPipe (Lugaresi et al., 2019).

Validation

All numbers come from model cards and result files in the repository, and `paper/make_figures.py` rebuilds every figure from them.

Galaxy CNN. On the 2,656-galaxy Galaxy10 validation split the model reaches 88.7 % ten-class, 97.2 % top-2 and 92.7 % coarse accuracy, against 82.8 %, 93.3 % and 89.5 % for ResNet-18; *Disturbed* (57 % recall) and *Unbarred loose spiral* (78 %) are the hardest classes (Figure 2). Accuracy on JWST and HST is expected to be lower because of the domain shift (Figure 3). Of three active-learning rounds, two challengers were correctly rejected and one was promoted (88.1 % Galaxy10 accuracy, 82 % recall on held-out stars and gas, 4.9 % of real galaxies rejected).

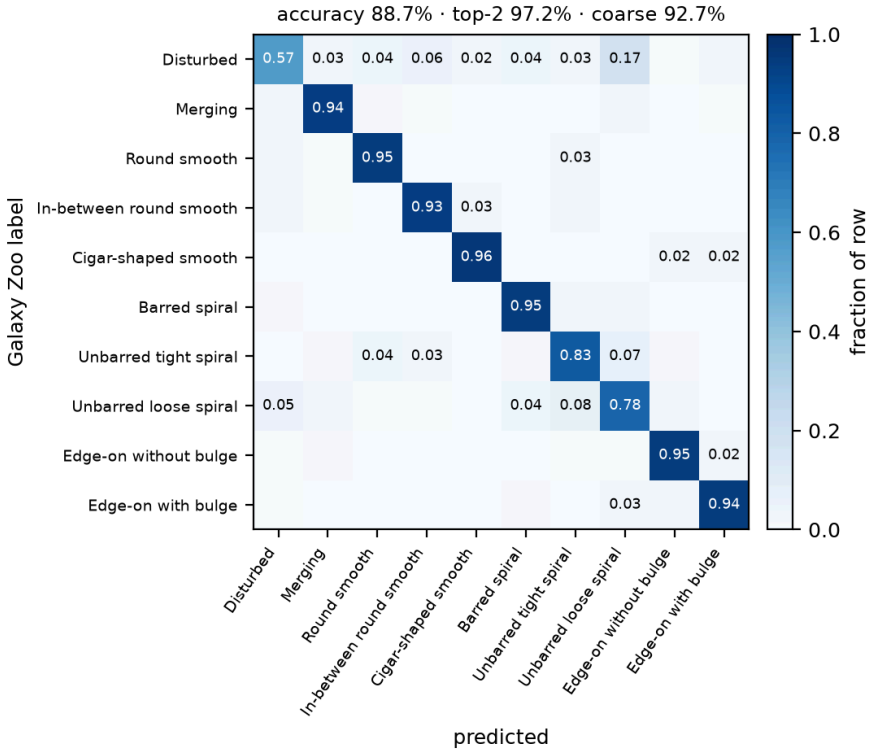


Figure 2: Galaxy CNN confusion matrix on the Galaxy10 DECaLS validation split, normalised by row.

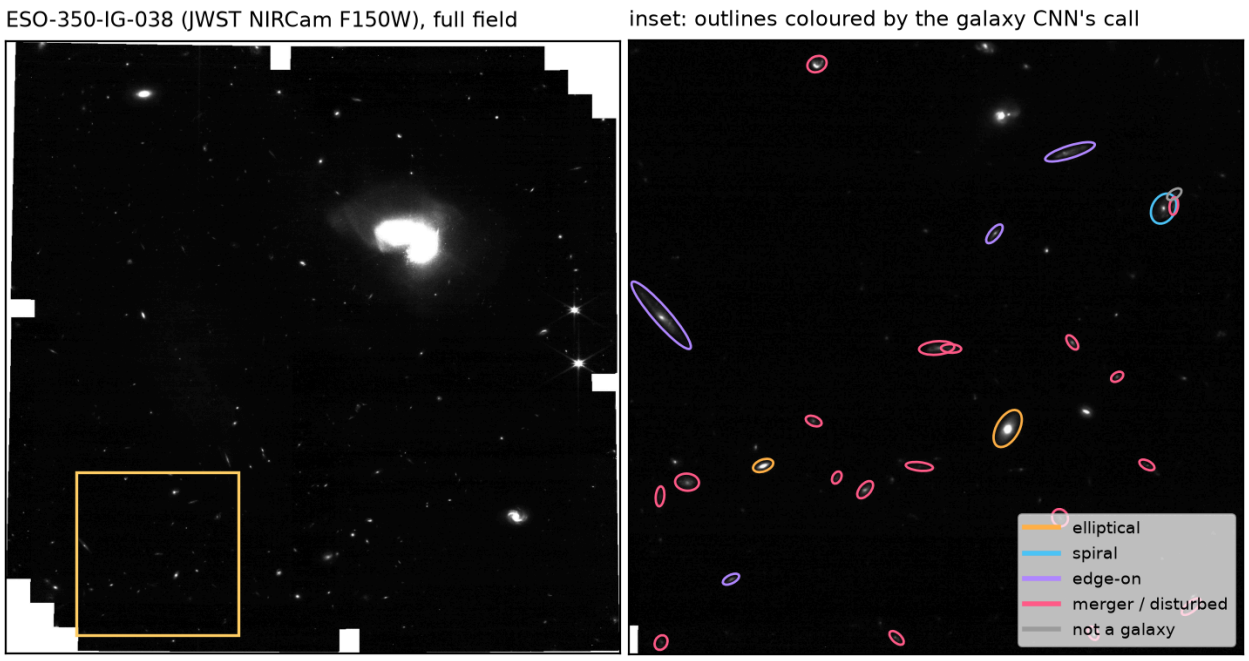


Figure 3: A public JWST NIRCam F150W image (ESO 350-IG038) processed by the pipeline. Right: the boxed region with each CNN-classified detection coloured by coarse class; the many merger/disturbed calls on small, faint sources show the domain shift from the training images.

Detection on calibrated data. On a JWST NIRCam F460M mosaic the detector found 12,029 sources with measured spurious rates of 0.24 % and 0.42 % (cold and hot passes); the low-surface-brightness passes failed the purity test and were dropped automatically. The 5σ point-source depth is AB 27.3, and 50 % completeness falls from AB 25.6 for the smallest injected galaxies to AB 21.7 for the largest (Figure 4).

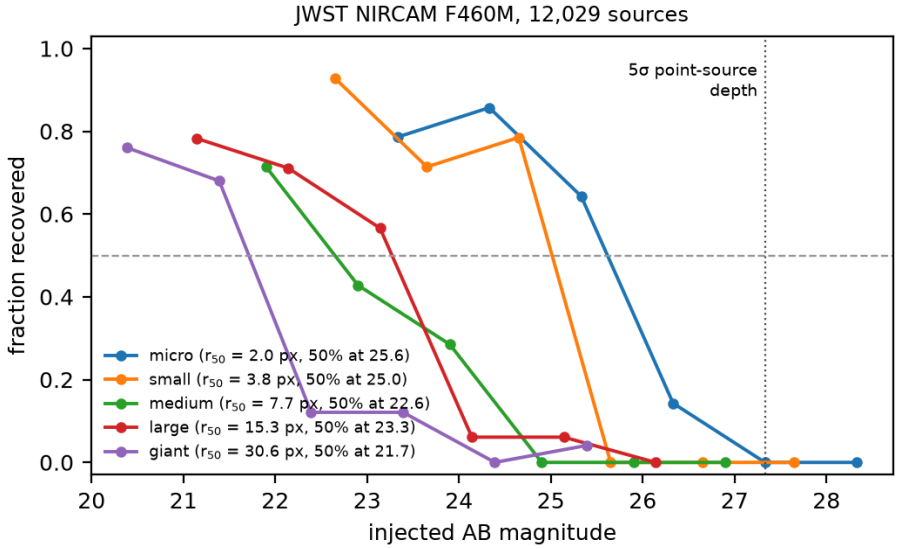


Figure 4: Injection-recovery completeness on a calibrated JWST NIRCam F460M mosaic, by size class (half-light radius r_{50}).

Cross-matching and Mars. Using Gaia DR3's Pan-STARRS1 best-neighbour table as an answer key, the matcher recovered 175 of 177 sources in 23 fields from Galactic latitude −74° to +76° (precision 98.9 %, recall 100 %). The Mars CNN reaches 80.0 % accuracy (89.0 % top-3, 61.1 % balanced) on the test split of Lu & Wagstaff (2020), which uses sols unseen in training.

Operation. In its first twelve days (30 October to 10 November 2026) the pipeline ingested 824 images, 1,870 Rubin and ZTF alerts and 46,455 detections, built 3,857 cross-survey objects from 29,191 archive matches, and raised 744 candidate anomalies.

Audit of anomaly candidates. The 744 candidates are not discoveries, and an audit of them shows why (Table 1). Only 394 concern individual astronomical objects; the rest are solar-activity changes or relays of events announced by other facilities, and only 56 (7.5 %) had been reviewed by a person. Of the 211 deep-field outliers, 96 are point sources and 26 more are rejected by the CNN as "not a galaxy", so the Isolation Forest mostly selects the brightest and largest sources. Of 29 broker tidal-disruption candidates, 23 were first detected two or more years earlier, which a flare lasting months cannot explain. Half of the "hostless" transients (8 of 16) are single-detection alerts that the broker classes as asteroids, and 29 of 57 supernova candidates carry an AGN, QSO or blazar class.

We therefore added vetting rules (`cosmic_watch/vetting.py`, with unit tests) that the pipeline and the re-measurement share: deep-field "normal" is defined over extended sources only; tidal-disruption candidates must be young and not identified as stars or AGN by the archives; a "new, hostless" verdict needs repeat detections, a successful check against known asteroids, and a recent first detection; and supernova candidates must not carry an AGN-type broker class. Re-applied to the same data (Table 1), the rules retracted 248 of 313 transient and outlier candidates. In a visual inspection of 24 randomly chosen outliers before and after, the fraction that were real galaxies rose from 9 of 24 to 14 of 24; the remaining contaminants are nebulae, satellite trails and detector artefacts. None of the surviving candidates has yet been confirmed as scientifically unusual. The next steps are adjudication against spectroscopic classifications from the Transient Name Server and a measured precision on a larger, independently vetted sample.

Candidate type	Raised	After vetting	Main reason for retraction
Deep-field outliers	211	33 (+51 newly selected)	star (96), CNN "not a galaxy" (28)
Tidal-disruption candidates	29	4	first detected years earlier (23)
Supernova candidates	57	28	broker class AGN/QSO/blazar (29)
"New, hostless" transients	16	0	broker asteroid (8), not new (6)
CNN mergers, nuclear transients	81	81	not yet vetted
Solar, relayed and other events	350	350	not object discovery
Total	**744**	**496 + 51**	56 had been reviewed by a person

Table 1: Anomaly candidates raised from 30 October to 10 November 2026, before and after the vetting rules.

Limitations. Classifications are probabilistic and outside the training domain; preview-image detections are indicative only; Rubin full frames are not public in real time. `Cosmic Watch` complements, and does not replace, the archives and brokers it draws on.

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